Addison Kopplin

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SUMMARY

Hands-on mechanical engineer who designs, builds, and debugs electromechanical hardware. Experience spans wire harness design for utility-scale solar projects, internal engineering tools and Excel-based automation, PCB design and embedded control, Simulink models tuned against real sensor data, and leading a high-power rocket team that designed an active roll control system. Thrives on getting into the weeds: debugging, troubleshooting, and figuring out how things work.

EXPERIENCE

Premier PV February 2024 – November 2025

Design Engineer I

- Designed wire harness systems for utility-scale solar projects, translating customer electrical drawings and CAD models into designs for manufacturing and installation
- Built internal engineering tools, including Excel-based automation, design templates, and work instructions, that improved workflow efficiency
- Ensured UL standards compliance and supported product development by coordinating with external testing laboratories
- Produced BOMs, engineering drawings, and design submittals for manufacturing and installation
- Partnered with procurement, production, and business teams to resolve supply chain issues, design changes, and manufacturing constraints

Milwaukee School of Engineering September 2022 – May 2023

Undergraduate Research Assistant

- Designed, assembled, and soldered a PCB that replaced a prototype breadboard-based system
- Tuned Simulink models and validated controller performance using collected accelerometer and encoder data
- Developed Simulink environments to streamline control, simulation, and system monitoring
- Implemented a vision-based pick-and-place task, using a webcam to detect and sort objects by color

ASME Design Competition September 2022 – May 2023

Controls & Electronics Engineer

- Designed and built the complete electronics system for a solar-powered robot on a 5-person engineering team
- Designed power and switching circuitry (voltage regulation, capacitor bank, MOSFETs, relays) to drive motors and actuators
- Implemented embedded control on an Arduino Nano with an integrated Bluetooth module for remote control
- Integrated and controlled an electromagnet for payload handling
- Hand-assembled and soldered all electronics on a prototyping board; supported system-level debugging and testing

AirPro Fan and Blower Company May 2022 – September 2022

Mechanical Engineer Intern

- Developed a bearing selection tool that helped engineers find bearings faster
- Verified blower performance through experiments measuring air speed in a duct
- Corrected SOLIDWORKS models and rebuilt assemblies to ensure high-quality product models and drawings

Rocket Design Competition September 2020 – May 2022

Team Lead

- Led design and fabrication of a high-power rocket for the Wisconsin Space Grant Consortium competition
- Designed an active roll control system using a reaction wheel and PID controller
- Simulated flight dynamics and analyzed flight data to drive design decisions
- Produced technical documentation and engineering reports

EDUCATION

Milwaukee School of Engineering

Bachelor of Science in Mechanical Engineering GPA 3.4

SKILLS

SolidWorks | AutoCAD | Inventor | MATLAB | Simulink | Python | C++ (Arduino) | Linux
Multimeter | Soldering | Assembly | Sensors & Data Acquisition | 3D Printing | PDM/ERP
Troubleshooting | Data Processing | Cross-functional Coordination | Engineering Documentation (creation/interpretation)